3D DESIGN AND VISUALIZATION OF A RESIDENTIAL ELECTRICAL WIRING SYSTEM

A Computer Graphics and Multimedia Project Proposal

Prepared by: Group 7

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1. Introduction

Electrical wiring is an important part of residential buildings. Electrical installations are commonly represented using two-dimensional drawings and electrical symbols. However, these drawings can sometimes be difficult to visualize, especially when trying to understand how the wiring is arranged within an actual building.

This project proposes the development of a **3D model of a residential house with its electrical wiring system**. The model will show basic electrical components such as lights, switches, sockets, a distribution board, and wiring within the house.

The 3D model will provide a simple and clear way of viewing the electrical installation and will allow the wiring to be observed through different views, including cutaway views where necessary.

2. Background

Computer-Aided Design (CAD) is widely used in engineering to design and visualize physical systems before they are constructed. Three-dimensional modeling provides a better representation of the physical arrangement of components compared to conventional two-dimensional drawings.

The group plans to use FreeCAD since it is a free and open-source CAD software that can be used to create accurate three-dimensional models. It provides a suitable platform for developing a simple residential building and representing its electrical wiring system.

This project will therefore combine basic electrical installation concepts with 3D CAD modeling to produce a visual representation of a residential electrical system.

3. Problem Statement

Residential electrical wiring is often represented using two-dimensional drawings, which may be difficult for some users to visualize in relation to the actual structure of a building. This makes it challenging to clearly understand the physical placement of electrical components and wiring routes.

There is therefore a need for a simple three-dimensional representation that can make residential electrical wiring easier to visualize and understand.

4. Proposed Solution

The project will develop a **3D residential house model using FreeCAD** and incorporate a basic electrical wiring system into the model.

The model will include:

- Electrical sockets

- Light fittings
- Switches
- Distribution board
- Basic wiring
- Electrical conduits where appropriate

The completed model will allow users to view the house and its electrical installation from different angles. Parts of the building can also be removed or made transparent to reveal the wiring inside the walls.

5. Main Objective

To develop a 3D model for visualizing a basic residential electrical wiring system using FreeCAD.

6. Specific Objectives

1. To design a simple residential house in FreeCAD.
2. To create 3D models of basic electrical components.
3. To place the electrical components within the house model.
4. To represent the basic wiring connections in 3D.
5. To produce clear views of the completed electrical installation.

7. Scope of the Project

The project will focus on creating a **simple residential house** and representing its basic electrical installation in three dimensions.

The house may include rooms such as:

- Living room
- Bedroom
- Kitchen
- Bathroom

The electrical system will include basic components such as:

- Lights
- Switches
- Socket outlets

- Distribution board
- Basic wiring

The project will focus on **3D modeling and visualization**. It will not involve physically wiring or energizing an actual house, nor will it involve complex electrical simulations or advanced electrical calculations.

8. Methodology

The project will be carried out through the following steps:

Step 1: House Design

A simple residential floor plan will be designed, showing the arrangement of rooms, doors and windows.

Step 2: 3D Modeling

The floor plan will be converted into a three-dimensional house model using FreeCAD.

Step 3: Electrical Component Modeling

Basic electrical components such as switches, sockets, lights and the distribution board will be created as 3D objects.

Step 4: Electrical Installation

The components will be positioned in appropriate locations within the house and connected using basic 3D wiring routes.

Step 5: Visualization

Different views of the completed model will be created. A cutaway or transparent-wall view will be used to show wiring that would normally be hidden inside the walls.

Step 6: Testing and Presentation

The model will be reviewed to ensure that the components are properly positioned and the wiring is clearly represented. The final model will then be presented and demonstrated.

9. Tools and Technologies

Software

- **FreeCAD** for 3D modeling and design
- Microsoft Office word for documentation
- Canva for making presentations.

Hardware

- Laptop or desktop computers
- Computer mouse and keyboard
- Storage device for project files

10. Division of Responsibilities

The project work will be divided among the five members as follows:

Member	Responsibility
Member 1	Design the house structure and floor plan
Member 2	Create the 3D house model in FreeCAD
Member 3	Create the 3D electrical components
Member 4	Add and arrange the wiring within the house model
Member 5	Combine the work, create final views and prepare the presentation

The responsibilities may be changed among members during the project depending on the group's needs. All members will participate in reviewing the model, documentation, testing, and final presentation.

11. Expected Results

At the end of the project, the group expects to produce:

1. A complete 3D model of a simple residential house.
2. 3D models of basic electrical components.
3. A basic electrical wiring system integrated into the house.
4. A cutaway or transparent view showing the wiring inside the building.
5. Final 3D views of the electrical installation.
6. A project report documenting the design and development process.
7. A demonstration of the completed model.

12. Significance of the Project

The project will provide a simple and visual way of understanding residential electrical wiring. Instead of viewing the installation only through conventional two-dimensional diagrams, users will be able to see how electrical components and wiring are positioned within an actual building.

The project will also give the group practical experience in:

- 3D computer-aided design
- Electrical system visualization
- Engineering design
- FreeCAD
- Teamwork and project development

The resulting model could also be used as a basic educational demonstration tool for students learning residential electrical installation.

13. Limitations

The project will have the following limitations:

- The model will represent a basic residential electrical system.
- It will not involve physical installation or energization.
- The project will not perform advanced electrical simulations.
- The model will focus mainly on visualization rather than complete professional electrical installation design.

14. Work Plan

Activity	**Time**
Project research and planning	Week 1
House floor plan design	Week 2
3D house modeling	Week 3
Electrical component modeling	Week 4
Wiring and component placement	Week 5
Model integration and corrections	Week 6

Final views and presentation preparation	Week 7
Report completion and submission	Week 8

15. Conclusion

This project proposes the development of a simple **3D residential electrical wiring model using FreeCAD**. The project will combine basic residential electrical concepts with 3D computer-aided design to create a clear visualization of how electrical components and wiring are arranged within a house.

The project is designed to be achievable within the available time and resources while providing the group with practical experience in engineering design, 3D modeling, electrical visualization, and teamwork.